

EKATA SINGH

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Education

Bachelor of Technology - Electronics and Communication Engineering	2022 - 2026
<i>Indira Gandhi Delhi Technical University For Women</i>	CGPA: 8.70
Senior Secondary (CBSE)	2021
<i>Kendriya Vidyalaya No. 2, Nau Sena Bagh</i>	95.4%
Secondary (CBSE)	2019
<i>Kendriya Vidyalaya No. 2, Nau Sena Bagh</i>	94.6%

Skills

Programming Languages: C++, Python, SQL.
Web Technologies Frameworks: HTML, CSS, JavaScript, React.js, Tailwind CSS, Langchain, LangGraph, VisionRAG
Developer Tools Others: Git | GitHub, GitLab.
CS Fundamentals: Data Structures and Algorithm | Object Oriented Programming | Database Management System | Operating System | Computer Networks | Information Retrieval

Experience

Nomura IT Intern (QA and Automation)	Jan'26 - Jun'26
- Built an AI-powered multi-agent system to automate SDLC release verification by validating Jira change requests, release tickets, and mandatory documentation against enterprise governance policies, reducing manual QA effort. - Developed an LLM-powered Test Case Generation Workbench from scratch that generates functional test cases from Jira Keys and Release Fix Versions, implementing Context Generation and Test Case Generation Agents using LangGraph workflows. - Engineered an enterprise RAG pipeline by ingesting project documentation from Jira and Confluence into a FAISS Vector Database, performing query expansion, dense sparse retrieval, document reranking, and Vision RAG to retrieve the most relevant contextual information for AI agents. - Built a React-based analytics dashboard to visualize workbench adoption across divisions, applications, and business units, and automated the IDATEN credit application using Playwright by developing reusable automation scripts, step definitions, and locator-based UI tests. Tech Stacks: Python, Flask, React.js, Playwright, LangChain, LangGraph, LLMs, AI Agents, Prompt Engineering, RAG Pipelines, Vision RAG, FAISS Vector Database, Dense Sparse Retrieval, Query Expansion, Document Reranking, REST APIs, Jira, Confluence, Git.	

Projects

RTL DeepPath Predictor Google Girl Hackathon 2025 GitHub	Feb'25 - Mar'25
- Developed an AI tool that cut RTL synthesis iterations by 70% and achieved 86.67% accuracy in predicting combinational logic depth within ± 1 level, addressing the challenge of lengthy hardware design cycles. - Improved model reliability by 18% in R^2 score (0.5119 to 0.8375) by expanding training dataset with 60+ complex RTL modules from 6 different hardware domains, resulting in more robust predictions across diverse applications.	
Memory Match Mania GitHub	Sep'23 - Dec'23
- Built a web-based card matching game with HTML, CSS, JS; used Fisher-Yates shuffle improving randomness by 20 - Improved performance with event delegation and dynamic DOM updates, cutting event listeners by 80%. Tech Stacks: HTML, CSS, JavaScript.	

Achievements

Selected **Amazon ML Summer School 2025** mentorship program.
Advanced to the **finals** of the prestigious **Google Girl Hackathon 2025**.
Semi-Finalist, **Flipkart GRiD 6.0** – Software Development Track
Leetcode – Solved 150+ problems.
Codeforces – Max Rating: 1658 — Expert

Extra-Curricular and Position of Responsibility

PR Head (TNP Cell, IGDТУW): Managed placement communication, engaging 20+ companies.	Apr'23 - Present
Mentor, Desh Ke Mentor: Mentored more than 10 high school girls.	Dec'22 - Apr'23